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HOMEOSTASIS

----- Active space --

Introduction

00:00:48

Homeostasis is the concept of constancy.

Central theme for bodily function.

Disturbance in homeostasis.

↓

Disease.

Claude Bernard :

Coined the term milieu interior (Internal environment).

Walter Cannon :

Coined the term homeostasis.

Control systems of homeostasis

00:05:48

Mechanisms of the body to maintain homeostasis.

Control systems

Negative feedback | Positive feedback | Feed forward

Negative feedback :

↑ in stimulus → ↓ parameters



Stimulus Response

Eg :

1. Baroreflex : Changes in BP

↑ during exercise ↓ after exercise.

2. Endocrine hormone regulation (99%).

3. Gastric acid secretion.

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Components of -ve feedback :

Stimulus : ↑BP
↓
Acts on receptors : Baroreceptors
↓
Afferents : Afferents
↓
Centre : Medulla
↓
Efferents : Efferent → SNS (−) / PNS (+) → ↓HR & ↓BP
↓
Response : BP

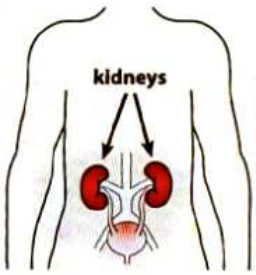
SNS : Sympathetic nervous system.
PNS : Parasympathetic nervous system.

Assessment of effectiveness of -ve feedback :

$$\text{Gain (G)} = \frac{\text{Correction by the System (C)}}{\text{Error remaining (E)}}$$

Infinite feedback gain :

When error becomes zero $G = \frac{C}{0} = \infty$
↓
Gain = Infinity





Eg :
Kidneys always correct blood pressure back to normal.
In HTN → Kidneys are affected → BP never comes back to normal.

Positive feedback :

- ↑ in initiating stimulus → Further ↑ in response.



- Amplification/cascade of reactions.
- AKA vicious cycle (Destabilizes our system).

Eg : CLAPS (mnemonic).

a) Coagulation cascade.

b) LH surge : Only exception for true feedback in endocrine hormone regulation.

c) Action potential.

d) Parturition (Normal delivery) :
AKA Ferguson reflex.

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Feedback

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Fetal head
↓
Pressure on uterine cervix ←
↓
Oxytocin release
↓
Uterine contraction
↓
Pushes fetal head further down

Cycle
↓
Eventually leads to delivery of fetus

e) Shock :

Hemorrhagic shock (Eg : RTA) → Irreversible shock → Death (+ve feedback).

Feedforward control system :

Core theme : Anticipatory control system.

Body responds anticipating a stimulus (No actual stimulus).

Eg :

a) Anticipatory tachycardia & tachypnea : ↑RR and ↑HR before exercise.

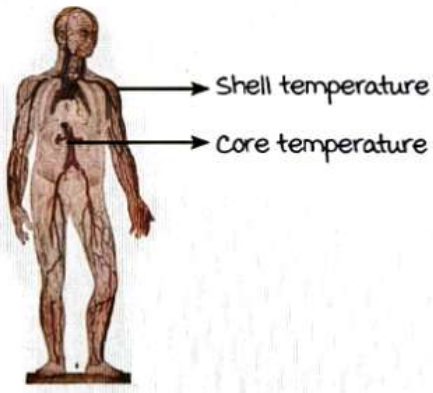
b) Anticipatory motor control :

- Centre : Cerebellum
- Person driving on a highway → Applies breaks on seeing a cow (Anticipates danger).

c) Anticipatory regulation of core body temperature :

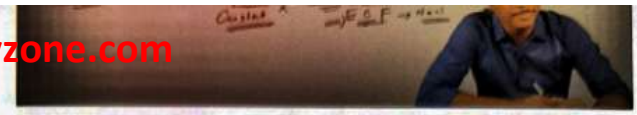
- Skin temperature → Shell temperature.
- Body organ temperature → Core temperature.

Cold environment.
↓
↓ shell temperature.
↓
Hypothalamus anticipates ↓ core temperature.
↓
Corrective measures.
↓
Maintanance of core temperature.



Feedback

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Properties of Homeostasis

00:28:04

Dynamic constancy :

AKA dynamic equilibrium.

Homeostatic control systems maintain a "range of values"

Eg : Normal HR = 60-100bpm.

Normal RR = 12-18bpm.



Prioritization of parameters :

- Priority of order of correction of parameters :

  pH > BP > Temperature.

- pH changes are given more priority : As they affect the activity of enzymes → Death.

Feedback